

Xinwen (Ellen) Zhang

Ph.D. Student at Temple University

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EDUCATION

Temple University

Ph.D. in Computer and Information Science

Advisor: Prof. Hongchang Gao

Research Interest: Designing large-scale optimization algorithms, including compositional optimization for AUC maximization, optimization beyond minimization, and optimization for large language models.

Tongji University

B.S. in Mathematics

Philadelphia, USA

Aug.2022 – May.2027

Shanghai, China

Sep.2017 – Jun.2021

AWARDS

- **Outstanding Graduate Research Assistant Award**, Department of Computer and Information Sciences, Temple University, 2026.
- **Female Student Award** in Data Science, IEEE International Conference on Data Mining, 2025.
- ICML 2026 Financial Aid Award, ICDM 2025 Student Travel Award, ICML 2024 Travel Award, WiML 2023 Travel Funding, NeurIPS 2023 Scholar Award, KDD 2023 Student Travel Award.
- Gold Reviewer, ICML 2026.
- Second Prize, Shanghai Mathematical Model Competition.
- Academic Scholarships, Three Consecutive Years at Tongji University.

RESEARCH EXPERIENCE

Ph.D. Research: Large-Scale Optimization Algorithms

Aug.2022 – Present

Advisor: Prof. Hongchang Gao

- **Compositional Optimization for AUC Maximization:**

Co-Advisor: Prof. Richard Souvenir

- Designing novel algorithms for compositional minimax optimization that progress from compositional to multi-level compositional formulations across single-machine and federated settings, employing variance-reduced and smoothing techniques to achieve state-of-the-art convergence rates.
- Establishing the connection between loss landscape geometry, optimization stability, and generalization for deep AUC maximization through sharpness-aware optimization.
- Applying the developed algorithms to diverse machine learning tasks, such as AUC maximization targeting imbalanced classification, and compositional optimization spanning portfolio optimization and multi-instance learning.
- **Optimization Beyond Minimization:**
 - Developing decentralized first-order algorithms for bilevel optimization over networks that eliminate the need for Jacobian/Hessian computation, substantially reducing communication overhead, and advancing this framework to establish robust convergence guarantees under heavy-tailed stochastic noise.
 - Investigating K -level optimization over networks, resolving the recursive dependency challenge in both algorithmic design and convergence analysis, employing compression techniques for communication efficiency, and building a unified theoretical framework for general K -level optimization.
 - Applying these algorithms to hyperparameter optimization and model pruning tasks, while relaxing the assumptions required on the lower-level function and supporting deeper multi-level hierarchies.
- **Optimization for Large Language Models:**
 - Exploring the convergence stability and robustness of the Muon optimizer for large-scale pretraining, investigating its behavior under heavy-tailed noise and in distributed training environments.
 - Investigating optimization algorithms for the post-training stage of large language models, including fine-tuning and knowledge distillation for efficient task-specific adaptation.

RESEARCH PUBLICATIONS

[ICDM 2026] Xinwen Zhang, Liang Zhan, Hongchang Gao. “Faster Federated Smoothed Minimax Optimization.” *In IEEE International Conference on Data Mining, 2026.*

[ICML 2026] Xinwen Zhang, Richard Souvenir, Hongchang Gao. “LS²MC-GDA: A Smoothed Algorithm for Federated Stochastic Multi-level Compositional Minimax Optimization.” *In Proceedings of the 43rd International Conference on Machine Learning, 2026.*

[ICML 2026] Xinwen Zhang, Yihan Zhang, Hongchang Gao, Heng Huang. “Distributed Stochastic K-Level Optimization Over Networks.” *In Proceedings of the 43rd International Conference on Machine Learning, 2026.*

[ICML 2026] Yihan Zhang, **Xinwen Zhang**, My T. Thai, Jie Wu, Hongchang Gao. “Convergence Analysis of Decentralized Hessian-/Jacobian-Free Algorithm for Nonconvex Stochastic Bilevel Optimization.” *In Proceedings of the 43rd International Conference on Machine Learning, 2026*.

[ToN 2026] Jiyao Liu, **Xinwen Zhang**, Xinliang Wei, Xuanzhang Liu, Yuzhou Chen, Hongchang Gao, Yu Wang. “Swapping and Purification Scheme Optimization for Entanglement Distribution in Quantum Networks.” *In IEEE Transactions on Networking, 2026*.

[NeurIPS 2025] **Xinwen Zhang**, Hongchang Gao. “On the Convergence of Stochastic Smoothed Multi-Level Compositional Gradient Descent Ascent.” *In Proceedings of the 39th Conference on Neural Information Processing Systems, 2025*.

[ICDM 2025] **Xinwen Zhang**, Hongchang Gao. “Sharpness-Aware Optimization Through Variance Suppression on Deep AUC Maximization.” *In IEEE International Conference on Data Mining, 2025*.

[IWQoS 2025] Jiyao Liu, **Xinwen Zhang**, Xinliang Wei, Xuanzhang Liu, Yuzhou Chen, Hongchang Gao, Yu Wang. “Joint Swapping and Purification with Failures for Entanglement Distribution in Quantum Networks.” *In IEEE/ACM 33rd International Symposium on Quality of Service, 2025*.

[ICML 2024] **Xinwen Zhang**, Ali Payani, Myungjin Lee, Richard Souvenir, Hongchang Gao. “A Federated Stochastic Multi-Level Compositional Minimax Algorithm for Deep AUC Maximization.” *In Proceedings of the 41st International Conference on Machine Learning, 2024*.

[NeurIPS 2023] **Xinwen Zhang**, Yihan Zhang, Tianbao Yang, Richard Souvenir, Hongchang Gao. “Federated Compositional Deep AUC Maximization.” *In Proceedings of the 37th Conference on Neural Information Processing Systems, 2023*.

Preprints

[arXiv 2025] **Xinwen Zhang**, Yihan Zhang, Heng Liang, Hongchang Gao. “Nonconvex Decentralized Stochastic Bilevel Optimization under Heavy-Tailed Noises.” *arXiv preprint arXiv:2509.15543*.

[arXiv 2025] **Xinwen Zhang**, Hongchang Gao. “On Provable Benefits of Muon in Federated Learning.” *arXiv preprint arXiv:2510.03866*.

[arXiv 2025] **Xinwen Zhang**, Hongchang Gao. “Federated Stochastic Minimax Optimization under Heavy-Tailed Noises.” *arXiv preprint arXiv:2511.04456*.

TUTORIALS & WORKSHOPS

[ICDM’25 Tutorial] Hongchang Gao, **Xinwen Zhang**. “Federated Stochastic Compositional and Bilevel Optimization.” *In the 2025 IEEE International Conference on Data Mining*.

[IJCAI’25 Tutorial] Hongchang Gao, **Xinwen Zhang**. “Federated Stochastic Compositional and Bilevel Optimization.” *In Proceedings of the 34th International Joint Conference on Artificial Intelligence, 2025*.

[WiML’23 Workshop] **Xinwen Zhang**, Richard Souvenir, Hongchang Gao. “A Fast Federated Stochastic Compositional Minimax Optimization Algorithm with Variance Reduction.” *18th Women in Machine Learning Workshop, 2023*.

[KDD’23 Tutorial] Hongchang Gao, **Xinwen Zhang**. “Distributed Optimization for Big Data Analytics: Beyond Minimization.” *In Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2023*.

WORK EXPERIENCE

Teaching Assistant at Temple University

Philadelphia, Aug.2022 – Dec.2024

- CIS3715 Principles of Data Science (Spring 2023, Spring 2024)
- CIS1051 Introduction to Problem Solving and Programming in Python (Fall 2022, Spring 2023, Fall 2023, Spring 2024, Fall 2024)

Data Analysis Intern at Zhongyuan Consumer Finance

Shanghai, Mar.2021 – Jun.2021

- Developed Python and SQL pipelines for automated credit monitoring and risk assessment, and built user profiling models from multi-source behavioral data to support targeted marketing strategies.
- Modeled behavioral patterns across advertising schedules and time periods to automate and improve the credit evaluation process.
- Conducted multivariate statistical analysis on user engagement data from a festival campaign, demonstrating that cash incentive activities significantly improved customer retention.

PROFESSIONAL SERVICE

• Conference Reviewer:

- 2026: AAAI, ICML, NeurIPS, KDD, ICLR, AISTATS.
- 2025: AAAI, ICML, NeurIPS, KDD, ICLR, AISTATS.
- 2024: AAAI, ICML, NeurIPS.

- **Journal Reviewer:** Journal of Combinatorial Optimization, Neural Networks, Journal of Optimization Theory and Applications.